

AMCS2104 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions like humans. To develop an AI system, four different approaches are implemented: Acting Humanly, Thinking Humanly, Thinking Rationally, and Acting Rationally.

- a) Briefly explain the **FOUR (4)** approaches. (8 marks)
- b) Identify another name for the Turing Test. (1 mark)
- c) To pass the Total Turing Test, a computer must possess certain capabilities. List all **SIX (6)** capabilities and explain their purposes. (12 marks)
- d) At any given time, rationality depends on four criteria to develop a rational agent. Briefly explain the **FOUR (4)** criteria. (4 marks)

[Total: 25 marks]

Question 2

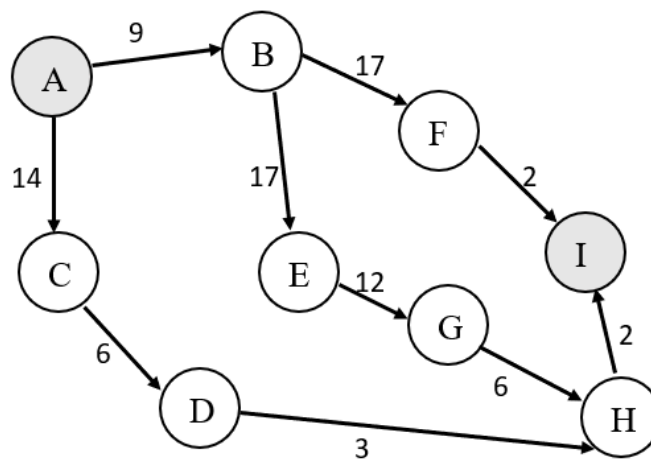
Hill climbing is an optimisation algorithm used to identify the best possible solution to a problem. It is commonly applied in AI and machine learning for local search optimisation.

- a) Hill climbing is also known as a greedy approach or greedy local search. Briefly explain why it is referred to in this way. (2 marks)
- b) Draw and label a state-space landscape in which its elevation corresponds to an objective function. (7 marks)
- c) Draw and label a state-space landscape in which its elevation corresponds to a cost function. (7 marks)
- d) Hill climbing search often gets stuck at a sub-optimal solution. Analyse the **THREE (3)** causes. (9 marks)

[Total: 25 marks]

Question 3

The Transportation Unit at TAR UMT has developed an AI-powered campus bus routing system designed to determine the shortest route from Station A to Station I. Figure 1 presents a directed graph consisting of nine campus bus stops, where the values indicate the travel distances (in kilometres) between each stop. Table 1 displays the heuristic cost of each stop relative to Station I, helping the system estimate the most efficient route.

**Figure 1.** Search Graph**Table 1:** Heuristic Cost of Each Station

Station	A	B	C	D	E	F	G	H	I
Euclidean distance	9	8	10	8	7	4	2	4	1

- Analyse the state space, initial state, goal state, action, transition model, and the cost function for the problem above. (6 marks)
- Based on the information provided in Figure 1 and Table 1, illustrate the resulting search tree for Breadth-First Search, Depth-First Search, and A* search in alphabetical order. (4 + 4 + 8 marks)
- Identify the solution returned by Breadth-First Search, Depth-First Search, and A* search in Question 3 b). (3 marks)

[Total: 25 marks]

Question 4

- a) A Sudoku is a popular puzzle game based on logical number placement. It has introduced millions to Constraint Satisfaction Problems (CSP), even if they do not realise it. A classical 6×6 Sudoku consists of 36 squares, some of which are initially filled with digits from 1 to 6. The objective is to fill the grid such that: Each row, column, and each classical 2×3 sub-grid contains all digits from 1 to 6, without repetition.
- (i) Formulate the 6×6 Sudoku as a CSP and identify the variables, domains for each variable, and the constraints using the **Alldiff** global constraint. (11 marks)
- (ii) A sample Sudoku puzzle is provided in Figure 2.

	1	2	3	4	5	6
A	3	1				2
B	4	6		1		
C				5		6
D	6				2	3
E			5			
F	2				6	4

Figure 2. Sudoku Puzzle

Suppose **Alldiff** constraints have been expanded into binary constraints so that the AC-3 algorithm (i.e., the most popular algorithm for enforcing arc consistency) can be applied directly. Explain how the following variables are made arc-consistent step-by-step.

- A4
 - C2
 - E6
- (9 marks)

- b) An Evolutionary Algorithm (EA) is a type of optimisation algorithm inspired by the process of natural evolution. It belongs to a broader class of algorithms known as evolutionary computation. These algorithms work by iteratively improving a population of candidate solutions using mechanisms that mimic biological evolution. Briefly explain the **FIVE (5)** key concepts in EA.

(5 marks)

[Total: 25 marks]